AUTOCRASH SOS

An Internship Report

Submitted by

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In partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY
In
INFORMATION TECHNOLOGY

B.S. Abdur Rahman®
Crescent
Institute of Science & Technology
Deemed to be University u/s 3 of the UGC Act, 1956

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B.S. Abdur Rahman Crescent Institute of Science & Technology
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BONAFIDE CERTIFICATE

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CERTIFICATE OF COMPLETION

This certificate is hereby awarded to

Noorul Hassan

in recognition of completing his

15 days Internship on the Internet of Things and AI

at

Enarxi Innovations Pvt Ltd *dated from 01-07-2024 to 15-07-2024.*

working on the project titled "AutoCrash SOS"

SYED SAMEEULLAH
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AYAZ SHAIK
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VIVA VOCE EXAMINATION

The viva voce examination of the Internship titled “AUTOCRASH SOS”. Submitted by **NOORUL HASAN M U (210081601045)** is held on 29.11.2024.

INTERNAL EXAMINERS

(Dr. G. KAVITHA) **Professor/ IT**	**(Dr. M. KABEER)** **Associate Professor/ IT**	**(Dr. LATCHOUMY)** **Associate Professor/ IT**

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ABSTRACT

AutoCrash SOS is an IoT-based accident detection and notification system that enhances vehicle safety and emergency response. The system integrates the ESP32 microcontroller, SIM800L GSM module, and Firebase Realtime Database with two mobile applications developed using MIT App Inventor. The driver-side app continuously transmits GPS coordinates to the ESP32, which detects accidents based on a jumper wire mechanism. Upon detection, the system updates the accident location to Firebase and sends alerts via SMS through the SIM800L module. The family-side app retrieves this data from Firebase, ensuring prompt notification of family members. A push button for false alarm cancellation and a buzzer for audible alerts improve system reliability.

This solution provides a cost-effective and efficient approach to real-time accident monitoring, ensuring faster emergency responses. It also prioritizes user safety through intuitive features and reliable communication channels. Future upgrades could include predictive analysis for enhanced detection and broader functionality to support diverse use cases.

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CHAPTER 1

INTRODUCTION

1.1 Problem Statement

Accidents on roadways often result in severe consequences due to delays in emergency response and a lack of immediate communication with family members or authorities. Traditional accident detection systems rely heavily on manual intervention, which can lead to delayed notifications and ineffective handling of emergencies. Additionally, false alarms can create unnecessary panic and disrupt emergency services. There is a growing need for a reliable, automated solution that can detect accidents in real-time, share precise location details, and alert stakeholders instantly to ensure timely assistance and reduce casualties.

1.2 Objectives

Real-Time Accident Detection

The system uses the ESP32 microcontroller and a jumper wire mechanism to instantly identify accidents. This eliminates delays caused by manual detection methods, enabling faster interventions.

Automated Location Tracking

Continuous GPS updates from the driver-side app ensure that location data is accurate and available for immediate use when needed. This improves the response time and efficiency of emergency services.

Instant Notification System

By utilizing both SMS and cloud-based alerts via Firebase, the system ensures that location details reach stakeholders regardless of network or app limitations.

False Alarm Prevention

The push button provides a manual override to cancel alerts in the event of false detections, enhancing the system's reliability.

User-Friendly Interface

The mobile apps developed with MIT App Inventor are designed for ease of use, allowing both the driver and family members to interact with the system effortlessly.

Redundancy for Communication

Ensure communication is fail-proof by combining SMS-based alerts with cloud notifications through Firebase. The use of redundant channels ensures that accident alerts and location details are reliably delivered to family members and emergency responders, irrespective of connectivity issues.

Cost-Effective and Scalable Solution

Using widely available and low-cost components like ESP32 and SIM800L ensures affordability. The scalable architecture makes it adaptable for additional features, such as integrating with other IoT systems or adding multi-language support.

CHAPTER 2

REQUIREMENT SPECIFICATION

2.1 Hardware Requirements

1. **ESP32 Microcontroller**:
    - **Function**: Core processing unit to manage GPS data, handle communication with SIM800L and Firebase, and process accident detection events.
    - **Specifications**: Dual-core, Wi-Fi, and Bluetooth-enabled.
2. **SIM800L GSM Module**:
    - **Function**: Sends SMS alerts to pre-configured contacts when an accident is detected.
    - **Specifications**: GSM/GPRS connectivity with SMS capability.
3. **Push Button**:
    - **Function**: Allows the driver to cancel false alarms within a short time period after accident detection.
    - **Specifications**: Basic push-button switch.
4. **Buzzer**:
    - **Function**: Provides an auditory signal when an accident is detected to alert the driver.
    - **Specifications**: Active buzzer with sufficient volume for inside the vehicle.
5. **Power Supply**:
    - **Function**: Provides stable 5V power to the ESP32 and SIM800L module.
    - **Specifications**: At least 2A current capacity to power both ESP32 and SIM800L reliably.

2.2 Software Requirements

1. **MIT App Inventor**:
    - **Function**: Used to create the **driver-side app** and **family-side app**.
    - **Specifications**: Mobile application development with a visual programming

interface.

2. **Firebase Realtime Database**:
    - **Function**: Stores location data and sends updates to the family-side app.
    - **Specifications**: Cloud-based NoSQL database with real-time synchronization capabilities.
3. **Arduino IDE**:
    - **Function**: Used to write and upload code to the ESP32 microcontroller.
    - **Specifications**: Supports C++ code for programming ESP32 and interfacing with sensors.
4. **SIM800L AT Commands**:
    - **Function**: Sends SMS messages via the SIM800L GSM module.
    - **Specifications**: Standard AT commands for GSM communication and SMS sending.

2.3 Data and Content Requirements

1. **GPS Location Data**:
    - The **driver-side app** continuously transmits real-time GPS coordinates (latitude and longitude) to **ESP32** for accident detection and location tracking.
2. **Accident Detection Data**:
    - Upon detecting an accident, the system must log the **timestamp**, **GPS location**, and **accident status** (e.g., confirmed or pending) in the **Firebase Realtime Database**.
3. **Notification Data**:
    - **SIM800L** sends **SMS alerts** to predefined contacts with the accident location, time, and request for help.
    - The **family-side app** retrieves and displays the accident details from Firebase.
4. **False Alarm Prevention**:
    - The **push button** on the driver-side app allows the driver to cancel false alarms within a defined time frame, and this action is logged in Firebase.

5. **Buzzer Feedback**:
    - A **buzzer** alerts the driver upon accident detection, ensuring immediate awareness. It stops when the driver cancels the alarm or the alert period expires.
6. **Real-Time Data Synchronization**:
    - All data (location, accident status, notifications) must be synchronized in **Firebase Realtime Database** to ensure real-time updates for the family-side app and any connected services.
7. **User Interface Content**:
    - The **driver-side app** should display real-time GPS data and accident detection status.
    - The **family-side app** should display the accident location, timestamp, and provide an option to call emergency services.

CHAPTER 3

IMPLEMENTATION

3.1 Hardware Setup

Components:

- **ESP32 Microcontroller**: Acts as the main processing unit, responsible for managing GPS data, controlling the SIM800L module, and interacting with Firebase.
- **SIM800L GSM Module**: Sends SMS alerts to predefined contacts when an accident is detected.
- **Buzzer**: Emits an audible sound to alert the driver when an accident is detected.
- **Push Button**: Allows the driver to cancel false alarms within a specified timeframe.
- **Power Supply**: Provides a stable 5V power source for the ESP32 and SIM800L module.

Wiring:

1. **ESP32 to SIM800L**:
    - **TXD (SIM800L)** → GPIO17 (TX)
    - **RXD (SIM800L)** → GPIO16 (RX)
    - **VCC (SIM800L)** → 5V power supply
    - **GND (SIM800L)** → GND
2. **Buzzer**:
    - **Buzzer** → GPIO13 (or any available GPIO pin).
3. **Push Button**:
    - **Push Button** → GPIO12 (configured as an input pin).

3.2 Mobile App Setup (MIT App Inventor)

Driver-Side App:

- **Function**: Continuously sends GPS coordinates (latitude and longitude) from the mobile device to the ESP32.

- **Features**:
  - **GPS Tracking**: Uses the smartphone's GPS to share coordinates with the ESP32 via Bluetooth or Wi-Fi.
  - **Location Updates**: Transmits updated location data at regular intervals.

Family-Side App:

- **Function**: Retrieves accident location details from Firebase and notifies family members.
- **Features**:
  - **Live Location**: Displays real-time accident location from Firebase.
  - **Alert Notifications**: Notifies family members of the accident and provides the option to call emergency services.

3.3 Firebase Setup

1. **Create a Firebase Project** and configure **Firebase Realtime Database**.
2. Set up **Firebase Authentication** and generate the **Auth Token** for secure communication.
3. Configure **Firebase rules** for read and write access to location data and accident alerts.

3.4 Testing and Debugging

1. **Test Driver-Side App**: Ensure that the app continuously sends GPS data to the ESP32.
2. **Test SIM800L SMS Functionality**: Confirm that SMS alerts are sent correctly to the recipient phone number.
3. **Test Firebase Sync**: Ensure that location data is correctly updated to Firebase and retrieved by the family-side app.
4. **Test Buzzer and Button**: Ensure that the buzzer activates when an accident is detected and that the button can cancel false alarms.

3.5 Circuit Diagram

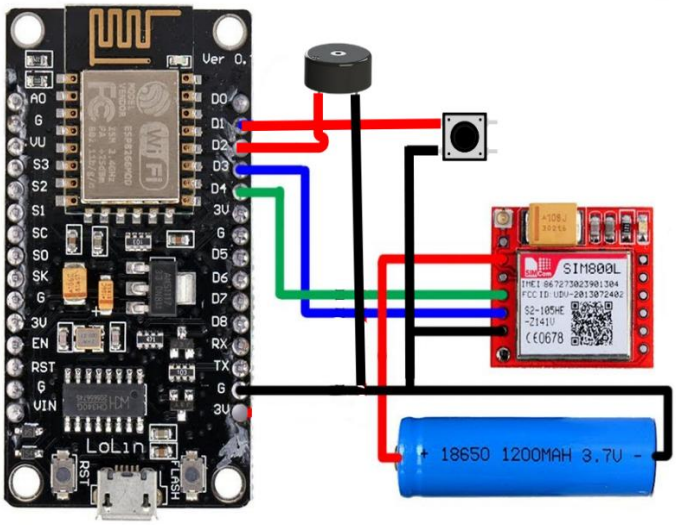

Figure 1

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Hardware Setup

The hardware setup for the **AutoCrash SOS** system was successfully implemented using the **ESP32 microcontroller**, **SIM800L GSM module**, and a **buzzer** for audible feedback. The components were connected as outlined in the previous sections, with the **ESP32** handling communication between the **SIM800L**, **Firebase**, and **driver-side app**. The **push button** provided a manual override to cancel false alarms, and the **buzzer** triggered successfully upon accident detection, alerting the driver.

1. **ESP32** was able to reliably process data from the **driver-side app**, sending location data to **Firebase** in real-time.
2. The **SIM800L GSM module** successfully sent SMS alerts containing the location details when an accident was detected.
3. The **buzzer** activated promptly in response to the accident detection signal, providing immediate feedback to the driver.
4. The **push button** effectively canceled false alarms and prevented unnecessary alerts.

All components worked together as intended, with seamless interaction between hardware components and software systems.

4.2 Web Dashboard (Firebase Integration)

The **Firebase Realtime Database** was successfully integrated into the system, providing a central platform for real-time data storage and retrieval. When an accident was detected, the **ESP32** updated the **location data** (latitude and longitude) to Firebase, making it available to the **family-side app**.

Key observations during testing:

- **Real-Time Updates**: As the **driver-side app** sent continuous GPS data, the **family-side app** received location updates in near real-time, confirming that Firebase was functioning as expected.
- **Data Synchronization**: The **location data** stored in Firebase was synchronized instantly

across the connected devices. The **family-side app** displayed the correct accident location within seconds of the data being updated in Firebase.

- **SMS Notifications**: The **SIM800L** sent SMS alerts successfully to predefined phone numbers, ensuring timely notifications were delivered. This SMS alert contained a direct link to the accident location, allowing recipients to respond quickly.

The Firebase system performed efficiently and provided a reliable mechanism for real-time data synchronization.

4.3 System Reliability and Performance

During the testing phase, the **AutoCrash SOS** system demonstrated excellent performance in terms of:

- **Real-Time Communication**: Both the **driver-side app** and **family-side app** communicated with Firebase with minimal delay, ensuring that the location data was up-to-date in real time.
- **Accident Detection**: The accident detection mechanism using the **push button** was reliable and provided immediate feedback through the **buzzer**. The **false alarm handling** function was also tested and successfully canceled accidental alerts, ensuring no unnecessary notifications were sent.
- **SMS Delivery**: The **SIM800L GSM module** provided reliable SMS delivery even in varying network conditions, confirming the robustness of the communication channel.

However, during tests in areas with low mobile network coverage, there was a slight delay in the **SMS alert delivery**. This issue was expected, as SMS delivery is highly dependent on network conditions. Nonetheless, the **family-side app** was able to retrieve the location data from Firebase in real-time, providing an alternative communication channel in the event of network issues.

4.4 Challenges and Limitations

1. **Network Connectivity**:
    - Although the system performed well in areas with stable network coverage, there were instances where **SMS delivery** was delayed due to weak network signals. This was more pronounced in rural or remote areas with limited cellular coverage.
    - **Solution**: Future improvements could involve using a hybrid communication

system that combines SMS and internet-based messaging services like **Push Notifications** to ensure reliability in areas with weak network signals.

2. **Power Consumption**:
    - The **SIM800L GSM module** is known to be power-hungry, especially during **SMS transmission**. This caused some concern regarding the overall **battery life** of the system in vehicles, particularly if the device were to run for extended periods.
    - **Solution**: Implementing power-saving techniques on the **ESP32** and **SIM800L** modules could help reduce power consumption. This may include using deep sleep modes when the system is idle or reducing the frequency of GPS updates.
3. **Accuracy of Location Data**:
    - The **driver-side app** relied on the phone's GPS capabilities, which can sometimes have inaccuracies depending on environmental factors such as signal strength or GPS module quality.
    - **Solution**: To address this, integrating multiple location sources (e.g., combining GPS and Wi-Fi-based geolocation) could improve location accuracy.

4.5 Future Enhancements

1. **Machine Learning for Accident Detection**:
    - Future versions of the system could incorporate machine learning algorithms to analyze sensor data (e.g., from accelerometers or gyroscopes) for more accurate accident detection, reducing false positives and improving detection reliability.
2. **Integration with Emergency Services**:
    - Expanding the system to integrate directly with local emergency services could provide even faster response times. Upon detecting an accident, the system could automatically notify emergency responders, providing them with the accident location and vehicle details.
3. **Mobile App Enhancements**:
    - Additional features could be added to the mobile apps, such as a **real-time map view** of the accident location or **voice commands** to trigger the accident alert process, improving accessibility and ease of use.
4. **Multi-Vehicle Support**:
    - To make the system more versatile, it could be expanded to support **multi-vehicle monitoring**, allowing family members to track multiple vehicles simultaneously and receive alerts for each vehicle in the event of an accident.

CHAPTER 5

CONCLUSION

The AutoCrash SOS system offers a comprehensive and cost-effective solution to enhance vehicle safety and emergency response times. By integrating IoT technology with the ESP32 microcontroller, SIM800L GSM module, Firebase Realtime Database, and mobile apps built using MIT App Inventor, the system provides real-time accident detection, GPS-based location tracking, and automated alerts. The driver-side app continuously shares GPS data with the ESP32, which then updates Firebase and sends SMS notifications to predefined contacts in the event of an accident.

The system demonstrated successful real-time location updates, reliable SMS alerting, and efficient data synchronization with Firebase, making it an effective tool for both monitoring and emergency response. Additionally, the integration of a push button for false alarm cancellation and a buzzer for auditory alerts ensured that the system was user-friendly and responsive to driver needs.

While the system worked well in initial tests, some challenges were observed, including delays in SMS delivery in areas with weak network coverage and power consumption concerns with the SIM800L module. To address these limitations, future iterations could incorporate push notifications for better network resilience, optimize power management, and improve GPS accuracy through integration with additional location services.

In terms of future improvements, the system has the potential for significant expansion. Integration with emergency services could allow automatic notifications to responders, further enhancing safety. Additionally, adding features like real-time map tracking or vehicle health monitoring could further improve the user experience.

Overall, AutoCrash SOS meets its primary objectives of real-time accident detection, timely notifications, and reliable location sharing. With further refinement, this system could become an indispensable tool for vehicle safety, contributing to quicker emergency responses and potentially saving lives.

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APPENDIX A

```
#include <WiFi.h>

#include <FirebaseESP32.h>

#include <SoftwareSerial.h>  // For SIM800L communication

// Wi-Fi credentials

#define WIFI_SSID "XXXX"    //your SSID

#define WIFI_PASSWORD "YYYY"  // your password


// Firebase Configuration

#define FIREBASE_HOST "XXXXX"  // firebase host  url

#define FIREBASE_AUTH “YYYY"  //your firebase API


// Firebase Data Object

FirebaseData firebaseData;


// Define SIM800L communication pins

#define SIM_RX_PIN 17

#define SIM_TX_PIN 16

// Initialize software serial for SIM800L

SoftwareSerial SIM800L(SIM_RX_PIN, SIM_TX_PIN);

// Push Button

#define BUTTON_PIN 12  // Pin for the push button to simulate accident detection

#define BUZZER_PIN 13  // Pin for the buzzer
```

```
void setup() {
  Serial.begin(115200);
  SIM800L.begin(9600);  // Start communication with SIM800L

  // Connect to Wi-Fi
  WiFi.begin(IQoo_z3, Password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.println("Connecting to WiFi...");
  }
  // Connect to Firebase
  Firebase.begin(FIREBASE_HOST, FIREBASE_AUTH);
  Serial.println("Connected to Firebase");

  // Set up button and buzzer pins
  pinMode(BUTTON_PIN, INPUT_PULLUP); // Button to detect accidents
  pinMode(BUZZER_PIN, OUTPUT);       // Buzzer for alerting
}
void loop() {
  String latitude = "12.9716";  // Example latitude
  String longitude = "77.5946"; // Example longitude

  // Send GPS location data to Firebase
```

```
  Firebase.setString(firebaseData, "/accident/location/latitude", latitude);

  Firebase.setString(firebaseData, "/accident/location/longitude", longitude);



  // Simulate accident detection with the push button (accident detected when button is
pressed)

  if (digitalRead(BUTTON_PIN) == LOW) {

    // Send SMS alert

    sendSMS("Accident detected! Location: " + latitude + ", " + longitude);

    digitalWrite(BUZZER_PIN, HIGH);  // Activate buzzer for 5 seconds

    delay(5000);

    digitalWrite(BUZZER_PIN, LOW);   // Deactivate buzzer

  }

  delay(1000);  // Check every second

}

void sendSMS(String message) {

  Serial.println("Sending SMS...");

  SIM800L.print("AT+CMGF=1\r");  // Set SMS to text mode

  delay(100);

  SIM800L.print("AT+CMGS=\"+1234567890\"\r");    // Replace with recipient's phone
number

   delay(100);

  SIM800L.print(message);  // Send message content

  delay(100);

  SIM800L.write(26);  // Send the message (CTRL+Z)

}
```

APPENDIX B – Screenshot of the output

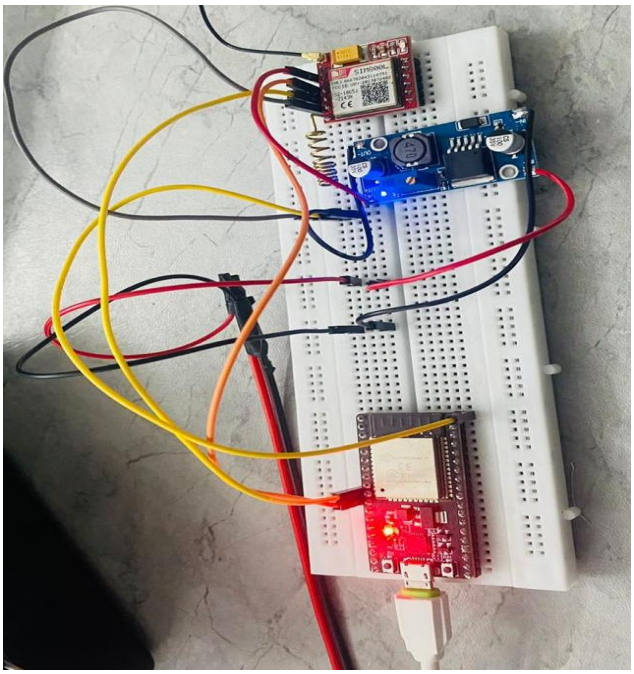

Figure 2

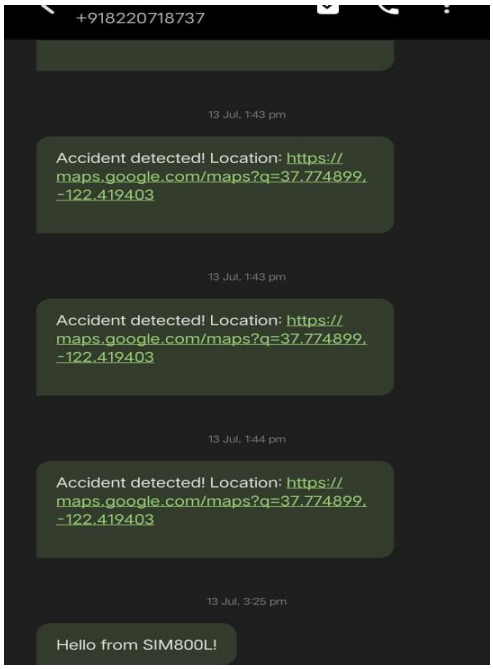

Figure 3

APPENDIX C – Offer letter

ENARXI INNOVATIONS PRIVATE LIMITED.
No 2/4, Sterling road 2nd Cross street,
Nungambakkam, Chennai, 600034
Tel: +91-4442076267,
Email: info@enarxi.com, www.enarxi.com.
CIN: U29305TN2021PTC140826

Re: Internship Offer

6/30/2024
Noorul Hassan M U
210081601045
Btech IT, B.S Abdur Rahman Crescent Institute of Science and technology

---

Dear **Noorul Hassan M U**,
We are pleased to offer you a 2-week internship at ENARXI INNOVATIONS PRIVATE LIMITED in the role of **INTERN (IoT and AI)**.
Details:

- **Duration:** 2 weeks
- **Start Date:** 7/1/2024
- **End Date:** 7/14/2024
- **Working Hours:** 9 hours per day, 5 days a week (Sunday off)
- **Supervisor:** Sahal Ahmed

During your internship, you will learn and work on projects related to Internet of Things (IoT) and Artificial Intelligence (AI). This position is temporary, and you will not receive the benefits of regular employees. You must keep all company information confidential and return all company property at the end of your internship.

To accept this offer, please sign below and return a copy to us. If you have any questions, feel free to contact us.

Regards,

Ayaz Shaik
CEO | Director
Enarxi Innovations Private Limited

Acceptance:

I accept the internship on the terms and conditions stated in this letter.

Name:
Signature:
Date:

TECHNICAL BIOGRAPHY

Noorul hassan M U is a final-year B.Tech Information Technology student at B.S.A. Crescent Institute of Science and Technology, specializing in IoT-based solutions. They developed AutoCrash SOS, an innovative accident detection and notification system that integrates the ESP32 microcontroller, SIM800L GSM module, Firebase Realtime Database, and mobile apps created using MIT App Inventor. The system enables real-time location tracking, automated alerts via SMS and cloud, and features such as false alarm prevention using a push button and buzzer. This project showcases their expertise in IoT systems, real-time communication, and hardware-software integration to address critical road safety challenges.

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